

NETWORK TRAFFIC SNIFFING

Packet Capture & RDP Traffic Analysis — Lab Report

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Report Type	Network Security / Packet Sniffing Lab
Tools Used	Wireshark, Windows Services (MMC), RDP (Port 3389)
Module	Sniffing
Result	Passed — 60% Score
Date	August 9, 2026

1. Objective

The objective of this exercise was to demonstrate practical packet sniffing and network traffic analysis using Wireshark. The lab focused on capturing and inspecting live traffic between two hosts on a local network, with particular attention to Remote Desktop Protocol (RDP) sessions carried over TCP port 3389, and on enabling the Windows Remote Packet Capture Protocol service to support remote packet capture.

2. Scope & Environment

Capture Tool	Wireshark
Capture Interface	eth0
Primary Hosts	10.10.1.13 and 10.10.1.3
Protocol of Interest	RDP (Fast-Path PDU) over TCP port 3389
Supporting Service	Remote Packet Capture Protocol v.0 (experimental)
Operating System	Windows (Services / MMC console)

3. Methodology

The following steps were carried out to complete the sniffing exercise:

- Enabled the "Remote Packet Capture Protocol v.0 (experimental)" service on the target Windows machine via the Services (Local) management console, allowing traffic on that machine to be captured remotely.
- Launched Wireshark and started a live capture on the network interface (eth0) connecting the two hosts.
- Observed general broadcast/ARP traffic to confirm host discovery, including an ARP request/reply resolving 10.10.1.11 to its MAC address.
- Filtered and followed the TCP conversation between 10.10.1.13 and 10.10.1.3 on port 3389 to isolate the RDP session.
- Analyzed the resulting Fast-Path PDU frames exchanged during the RDP session, noting sequence numbers, acknowledgements, and window sizes.
- Completed the accompanying "Sniffing" assessment module to validate understanding of the capture and analysis process.

4. Findings

4.1 Remote Packet Capture Protocol Service

The "Remote Packet Capture Protocol v.0 (experimental)" service was located in the Windows Services console and confirmed to be running with a Manual startup type under the Local System account. This service allows traffic on the host to be captured from a remote Wireshark instance, which is a prerequisite for the remote-capture portion of the exercise.

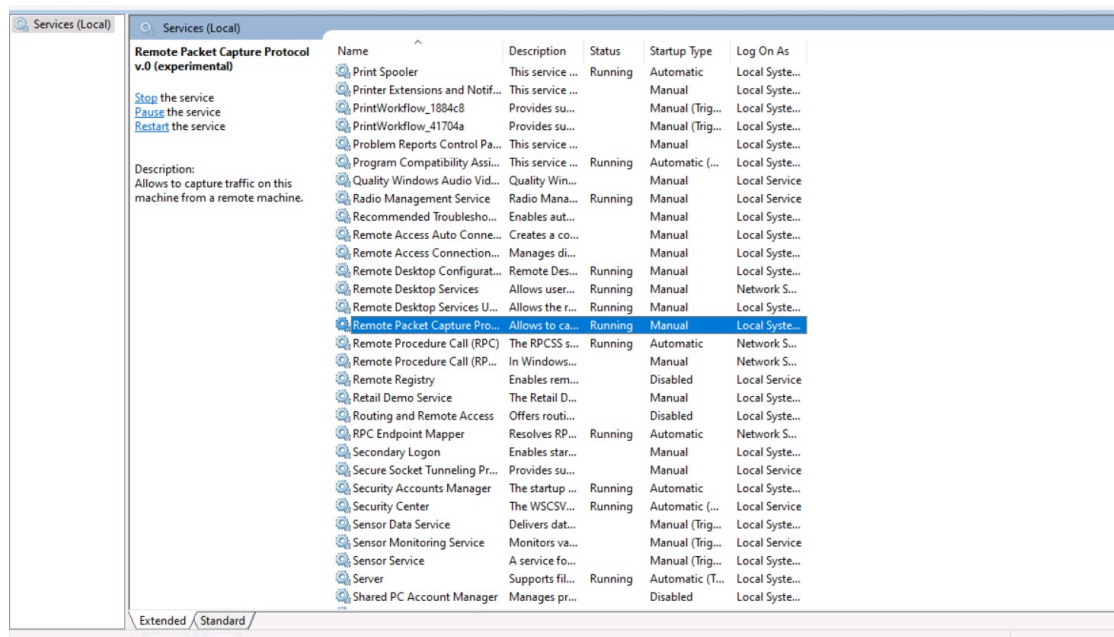


Figure 1: Remote Packet Capture Protocol service shown as Running in Windows Services (Local).

4.2 General Traffic Capture

An initial capture showed a mix of background IPv4 traffic alongside the TCP conversation of interest. An ARP exchange was also observed, in which 10.10.1.13 requested the MAC address for 10.10.1.11, which was resolved as belonging to Microsoft_01:80:11. Multiple PSH/ACK segments were exchanged between 10.10.1.13 and 10.10.1.3 on port 3389, consistent with an active RDP session transferring data.

No.	Time	Source	Destination	Protocol	Length	Info
3942	32.380773821	10.10.1.13	10.10.1.3	TCP	98	3389 → 42382 [PSH, ACK] Seq=11220238 Ack=10775 Win=551 Len=52 TSval=4073
3943	32.381045139	10.10.1.3	10.10.1.13	TCP	66	42382 → 3389 [ACK] Seq=10775 Ack=11220270 Win=25854 Len=0 TSval=22308062
3944	32.389451795	190.62.211.124	21.200.5.73	IPv4	54	
3945	32.389697911	90.132.39.105	101.142.253.19	IPv4	54	
3946	32.389743314	98.156.61.45	233.192.136.84	IPv4	54	
3947	32.389787417	83.39.4.95	231.113.251.60	IPv4	54	
3948	32.389866322	183.106.412.93	200.221.127.50	IPv4	54	
3949	32.389910625	161.190.207.64	251.48.155.35	IPv4	54	
3950	32.389952128	174.57.13.93	77.96.24.89	IPv4	54	
3951	32.389988830	48.134.142.31	217.100.21.35	IPv4	54	
3952	32.390041734	115.229.161.67	221.253.49.26	IPv4	54	
3953	32.390088537	94.22.28.17	8.184.161.93	IPv4	54	
3954	32.425351467	10.10.1.13	10.10.1.3	TCP	4169	3389 → 42382 [PSH, ACK] Seq=11220270 Ack=10775 Win=551 Len=4103 TSval=4073
3955	32.425645886	10.10.1.3	10.10.1.13	TCP	66	42382 → 3389 [ACK] Seq=10775 Ack=11224373 Win=25822 Len=0 TSval=22308063
3956	32.440686680	MS-NLB-PhysServer-2	Microsoft_01:80:11	ARP	42	Who has 10.10.1.11? Tell 10.10.1.13
3957	32.441997387	Microsoft_01:80:11	MS-NLB-PhysServer-2	ARP	42	10.10.1.11 is at 00:15:5d:01:80:11
3958	32.478983511	10.10.1.13	10.10.1.3	TCP	7306	3389 → 42382 [PSH, ACK] Seq=11224373 Ack=10775 Win=551 Len=7240 TSval=4073
3959	32.478993311	10.10.1.13	10.10.1.3	TCP	3609	3389 → 42382 [PSH, ACK] Seq=11231613 Ack=10775 Win=551 Len=3543 TSval=4073
3960	32.479310332	10.10.1.13	10.10.1.3	TCP	2962	3389 → 42382 [PSH, ACK] Seq=11235156 Ack=10775 Win=551 Len=2896 TSval=4073
3961	32.479829966	10.10.1.13	10.10.1.13	TCP	66	42382 → 3389 [ACK] Seq=10775 Ack=11231613 Win=27438 Len=0 TSval=22308063
3962	32.479830166	10.10.1.13	10.10.1.13	TCP	66	42382 → 3389 [ACK] Seq=10775 Ack=11235156 Win=27411 Len=0 TSval=22308063
3963	32.479854768	10.10.1.13	10.10.1.3	TCP	10202	3389 → 42382 [PSH, ACK] Seq=11238052 Ack=10775 Win=551 Len=10136 TSval=4073
3964	32.479959575	10.10.1.13	10.10.1.13	TCP	66	42382 → 3389 [ACK] Seq=10775 Ack=11238052 Win=27389 Len=0 TSval=22308063
3965	32.479967876	10.10.1.13	10.10.1.3	TCP	4410	3389 → 42382 [PSH, ACK] Seq=11248188 Ack=10775 Win=551 Len=4344 TSval=4073
3966	32.479973976	10.10.1.13	10.10.1.3	TCP	11427	3389 → 42382 [PSH, ACK] Seq=11252532 Ack=10775 Win=551 Len=11361 TSval=4073

Figure 2: Wireshark capture showing TCP/ARP traffic and the RDP conversation on port 3389.

4.3 RDP Fast-Path PDU Analysis

A focused filter on the RDP conversation revealed a continuous stream of Fast-Path PDU frames (NewPointer, ColorPointer, CachedPointer, PointerPosition, LargePointer, Palette, Synchronize, and Surface types), interspersed with TCP ACK segments. Several of these PDUs were flagged by Wireshark as "Malformed Packet," and one TCP segment was identified as a retransmission (frame 7349), indicating some packet loss or dissection limitations during the capture. The Ethernet frame details confirmed the source and destination MAC addresses for the two NLB-clustered physical servers involved in the exchange.

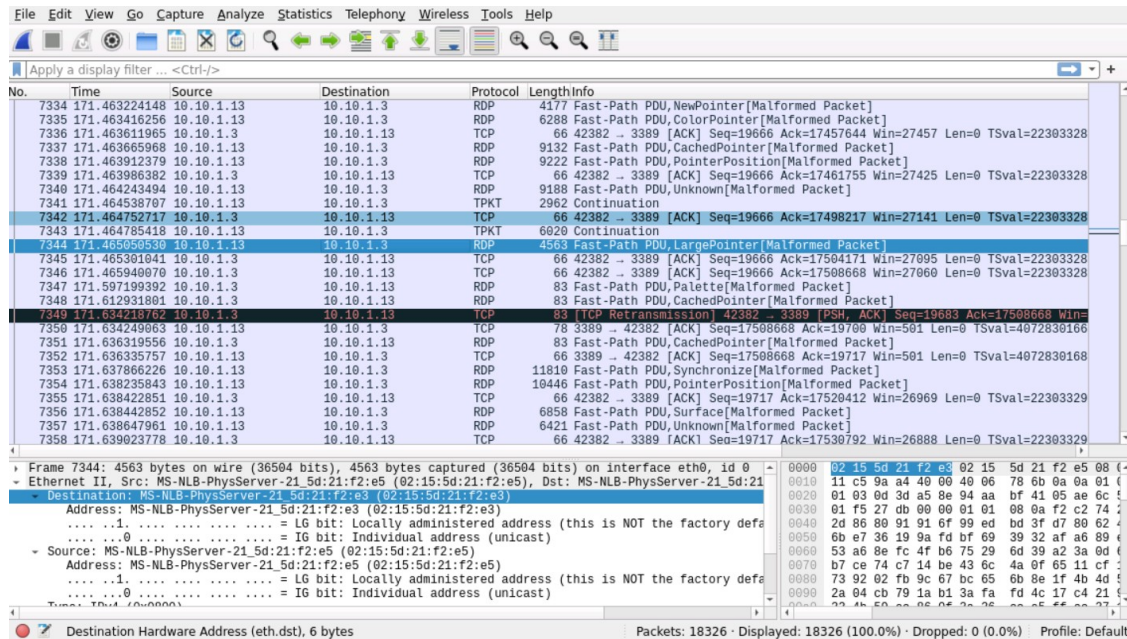


Figure 3: Detailed view of RDP Fast-Path PDU frames, including a TCP retransmission and Ethernet frame breakdown.

5. Assessment Result

Upon completion of the practical capture and analysis steps, the associated "Sniffing" module assessment was submitted and passed with a score of 60%. The task was completed in 2 hours and 1 minute against an expected duration of 1 hour and 20 minutes.

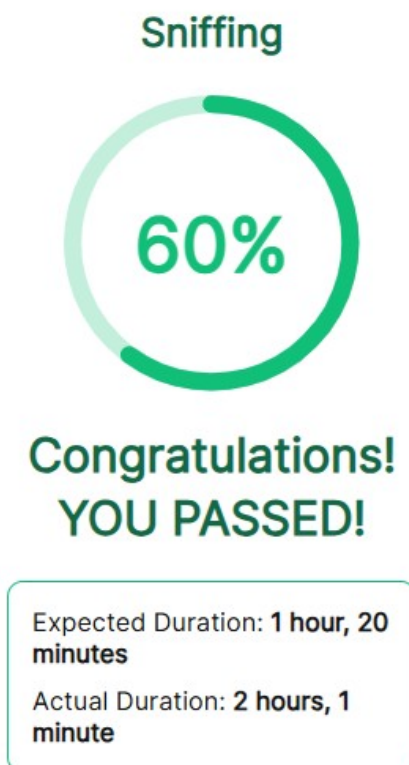


Figure 4: Module completion summary — Sniffing module passed at 60%.

6. Conclusion

This exercise provided hands-on experience with network sniffing fundamentals: enabling a remote capture service, capturing live traffic with Wireshark, and dissecting an RDP session at the packet level. The presence of malformed Fast-Path PDUs and a TCP retransmission highlighted the kind of anomalies that appear in real-world captures and underscored the importance of careful frame-by-frame analysis when investigating network traffic. The exercise was completed successfully, with the associated assessment module passed.

7. Recommendations

- Disable the Remote Packet Capture Protocol service when not actively in use, as it increases the remote attack surface if left running.
- Where possible, use encrypted RDP configurations (e.g., Network Level Authentication and TLS) to reduce the value of intercepted traffic to an eavesdropper.
- Investigate the cause of the malformed PDUs and the observed retransmission to rule out network instability or an MTU/fragmentation issue.
- Revisit the areas that led to the 60% score to reinforce weaker concepts before attempting more advanced sniffing or protocol-analysis modules.